

Nudging for Online Misinformation: a Design Inquiry

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ABSTRACT

The use of technology to combat online misinformation is becoming a prominent topic of research and practice, yet most of the existing solutions do not consider the impact of cognitive biases in misinformation-related decision-making. Addressing this gap, the present paper explores the potential of *nudging* to inform the design of technological interventions that combat misinformation. We report on a design workshop with 29 participants, who were asked to conceive technology-mediated nudges for misinformation with the use of the "Nudge Deck", a design-support tool that presents 23 interaction design mechanisms for nudging.

CCS CONCEPTS

• **Human-centered computing** → **HCI theory, concepts and models.**

KEYWORDS

nudging; online misinformation; design workshop; behavior change

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1 INTRODUCTION

Misinformation, defined as inaccurate or misleading information that is spread unintentionally [31], has shown to have the capacity to shape public opinion [11], to harm democratic procedures [11], to ignite violence with real-life consequences [25] and to create a breeding ground for harmful conspiracy theories to flourish [20], among others. Online environments have made both the study, as well as the propagation of misinformation, more efficient. Recent studies, for instance, have shown that online misinformation propagates faster and further than true information [31]. On Twitter, the top 1% of misinformation diffuses to 1-10K users, while true news hardly reaches an audience greater than 1K. Misinformative news is more likely to be retweeted than true stories, with the probability

being as high as 70% [31]. It is estimated that 25-30% of social media users to have shared misinformation, knowingly or not [1, 2].

Interest in technological interventions to mitigate misinformation has grown rapidly in recent years. The vast majority of work has focused on spotting misinformation, leveraging algorithmic classification techniques (e.g., [33]), crowdsourcing (e.g., [32]), as well as information aggregation from various fact-checking databases (e.g., [15]). Less thought has been spent on how information should be presented to users, once a likely engagement with misinformation is detected. For instance, Fernandez and Alani [10] highlighted that most interventions do not go beyond the simple presentation of facts. Leetaru [18] calls this the *magic bullet concept*: simply presenting corrective information in hopes of changing individuals' behavior. Such an approach can even backfire, as presenting corrective information has been repeatedly found to fail in changing individuals' behavior and often, to reinforce pre-existing views [12].

In this work-in-progress, we explore the potential of *nudging* [27] to guide the design of user interfaces that combat misinformation. The underlying assumption is that online information consumption is dominated by shallow cognitive processing [21]. Online information consumers are cognitive misers, reluctant to engage in deliberative and effortful information processing, which makes them susceptible to a plethora of *cognitive biases*, such as the *confirmation bias* (i.e., favoring information that confirms pre-existing beliefs) [22], the *bandwagon effect* (i.e., adopting behaviors, actions, or attitudes simply because others are doing so) [24], the *illusory-truth effect* (i.e., repeated exposure of false information increases its believability) [23] or the *novelty effect* (i.e., being more attracted to or influenced by new information) [31].

We report on a design workshop, in which 29 participants were asked to use the "Nudge Deck" [5] in designing technology-mediated nudges against misinformation. We briefly describe the concepts that emerged from the workshop, mapped onto three stages of the online news process framework (i.e., *composition*, *amplification*, and *consumption*) [28] and outline our future work.

2 BACKGROUND

2.1 Judging news veracity: A dual-process perspective

Dual-processing theories of decision-making suggest that cognition can be divided in two categories: System 1 and System 2 [14]. System 1 refers to automatic, impulsive and instinctive processes, while System 2 focuses on slow, reflective, time-consuming thinking [9]. Humans have a disposition of reducing effort, thus relying on System 1 and only applying System 2 in cases where the automatic system cannot manage [17]. Neuroscience research suggests

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that System 1 accounts for up to 95% of individuals' daily cognitive activities and decision-making [30]. System 1 relies on mental shortcuts (*i.e.*, heuristics) which allow individuals to quickly reach reasonable conclusions or solutions to complex problems and avoid cognitive overload [26]. For instance, when individuals are uncertain on whether a news article is misinformative or not, they will attempt to form a judgement by anchoring to a salient and important value (*e.g.*, the source of the article) and use it as a reference for making adjustments [26]. However, heuristics can result in errors and cause individuals to draw incorrect conclusions. Research in the field of Behavioral Economics has unveiled a number of ways in which humans make systematic decision errors, the so-called *cognitive biases* [17]. One example is the *anchoring bias*, individuals' tendency to heavily rely on an anchor (*e.g.*, the source of the information, as demonstrated in the example above) and ignore any other available information, thus skewing their judgment [19].

2.2 Nudging

Thaler and Sunstein [27] introduced the notion of nudging to suggest that knowledge about cognitive biases can be leveraged through the environment in which people make choices. They define nudging as *"any aspect of the choice architecture that alters people's behavior in a predictable way, without forbidding any options or significantly changing their economic incentives"* [27].

2.3 The "23 Ways to Nudge" framework

Translating, however, a cognitive bias into a nudge is not a trivial process. Cognitive biases, we argue, tell us about the *"why"* of nudging - what the problem we are trying to solve is. As interaction designers, we lack knowledge on the *"how"* of nudging - what exact interaction mechanisms we can employ to steer individuals' behaviors. To address this problem, we created the *"23 Ways to Nudge"* framework [4]. The framework came out of a systematic review of 71 papers published in the Human-Computer Interaction (HCI) space, which proposed novel technology-mediated nudging solutions. The framework identifies 23 distinct nudging mechanisms, clustered into six categories, leveraging 15 different heuristics or cognitive biases [4]. The six nudge categories are: *Facilitate* (reducing physical or mental effort required for an activity), *Confront* (pausing unwanted - mindless actions by introducing doubt), *Deceive* (affecting how options are presented and how activities are experienced), *Social Influence* (exploiting individuals' eagerness to comply with social norms and expectations), *Reinforce* (promoting behaviors by increasing their salience in individuals' mind) and *Fear* (provoking feelings of anxiety, loss and doubt to promote a course of action).

3 METHOD

With the goal of exploring the design space of technology-mediated nudging for misinformation mitigation, we conducted a design workshop, in which 29 participants were asked to use the *"Nudge Deck"* in conceiving novel ways to change users' misinformation-related behaviors, such as dissuading them from undesired behaviors like posting, sharing or consuming misinformation and promoting desirable behaviors, such as engaging with or sharing fact-checked information.

3.1 The "Nudge Deck"

The *"Nudge Deck"* is a card-based, design support tool for the implementation of nudging interventions [5]. The *"Nudge Deck"* was developed to make the *"23 Ways to Nudge"* framework [4] accessible to design teams during time-constrained meetings. It consists of 32 double-sided cards: 23 nudging mechanism cards outlining the nudge mechanisms along with illustrated examples, as well as design considerations that help generate discussion during the design meeting, six category cards that are color-coded to simplify their recognition and three trigger cards: *facilitators*, *sparks*, and *signals*. The trigger cards are based on Fogg's Behavior Model [13], indicating the type of nudge that is most suitable in a given context: *facilitator* nudges increasing one's ability to perform a behavior, *spark* nudges boosting one's motivation to perform a behavior and *signal* nudges reminding users to perform a behavior [4]. Each nudge mechanism card is mapped to one trigger card and bears on the card's front side the associated trigger symbol.

3.2 Participants

A total of 29 participants joined the study (14 male; 15 female; median age=29). Participants were graduate students in an HCI-related field, at the PhD or Master's level, while most of them had prior experience with behavior change concepts.

3.3 Procedure

Participants were grouped into teams of three or four members. They were asked to review a definition of nudging along with examples, as well as the *"Nudge Deck"*. Then, they received a design brief (see A.1) and templates (see A.2), on which they were asked to sketch out their ideas along with short descriptions and annotations. The design activity lasted 45 to 60 minutes. At the end of it, participants were asked to present their produced ideas and their design rationale. All activities were audio or video recorded with participants' consent.

3.4 Analysis of design concepts

The analysis of the generated nudges consisted of two steps: *sketch refinement* and *conceptualization*. In the first step, we turned participants' hand-drawn sketches into low and mid-fidelity wireframes with sufficient details and visual refinements, eliminating possible ambiguities in the designs. Once completed, we conversed with the design teams and compared their own sketches to the new wireframes for resolving possible differences. In the second step, we aimed to gain a deeper understanding of the design space of the generated nudges for combating online misinformation. For doing so, we followed an iterative analysis process, comparing, expanding and mapping ideas back to the *"23 Ways to Nudge"* framework taxonomy [4]. We expanded upon the ideas by specifying or altering the nudge mechanisms, or re-appropriating them for different contexts, while in other cases, we re-appropriated the ideas to better fit principles of nudging. Last, we assigned a name to each concept, reflecting its strategy for behavior change (*e.g.*, *Blurry* mitigates misinformation amplification through a *blur effect*).

4 FINDINGS

All in all, the design workshop resulted in eighteen technology-mediated interventions combating online misinformation. Fifteen concepts were based on the nudge mechanisms provided by the "23 Ways to Nudge" framework [4] and three relied on other behavior change theories, introduced by participants themselves, leveraging concepts from the theories of *Social Translucence* [8], *Inoculation* [29] and the *Continued Influence Effect* [6].

We briefly present below the concepts that emerged from the workshop, mapped onto three stages of the online news process framework, as conceived by Turk [28]: *composition*, *amplification* and *consumption*. This framework hints different intervention approaches for combating misinformation; in the *composition* stage, researchers explore ways to discourage the creation of misinformation, in the stage of *amplification*, they come up with ways to minimize its spread and for *consumption*, they encourage users to be aware of the quality of information they consume [28]. We attempted to group all generated concepts to these stages, based on their behavioral goal (*i.e.*, the course of action prompted: mitigate the creation, amplification or consumption of misinformation).

4.1 Composition stage

Four nudges targeted the *composition* stage (see table 1), attempting to reduce the likelihood of posting misinformation on social media. Three of the nudges do so by *instilling doubt* (see [4] for a description of the 23 nudging mechanisms), while another exploits users' need to conform to social norms and expectations.

Table 1: Nudges combating the composition of misinformation

Description	Bias (Mechanism)
<i>Await</i> adds a 10-sec buffer to the post button when a non-credible link is included in a post.	Regret-aversion bias (Throttle mindless activity)
<i>Aureole</i> analyzes links in a post and provides colored feedback, along with a short message (see fig. 1).	Regret-aversion bias (Create friction)
<i>Noticeable</i> modifies the exposure of posts, depending on the outcome of a fact-checking analysis.	Regret-aversion bias (Create friction)
<i>Socially</i> highlights users in one's network that are likely to find a generated post as misinformative.	Spotlight effect (Raising visibility of user actions)

4.2 Amplification stage

Four nudges targeted the *amplification* stage (see table 2), aiming to minimize users' interactions with misinformation (*e.g.*, sharing, liking, commenting).

4.3 Consumption stage

Ten concepts targeted the *consumption* stage (see table 3), aiming to advance users' critical skills.

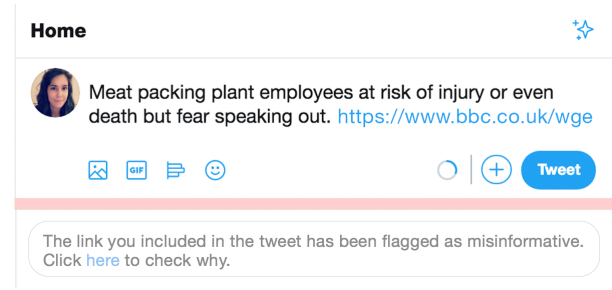


Figure 1: Aureole notifies the user that the attached link is non-credible

Table 2: Nudges combating the amplification of misinformation

Description	Bias (Mechanism)
<i>Blurry</i> applies a masked effect to posts containing non-credible links. Users are able to check the posts, opening them in a new tab.	Status-quo bias (Default)
<i>Name It</i> issues a message when users share non-credible posts, warning that such posts are shared by non-credible accounts.	Herd-instinct bias (Enabling social comparisons)
<i>Changy</i> recommends users to switch to a fact-checked article when they are sharing non-credible posts.	Authority bias (Provide multiple viewpoints)
<i>Alternative</i> suggests a fact-checked article for examination when users are sharing non-credible posts.	Confirmation effect (Provide multiple viewpoints)

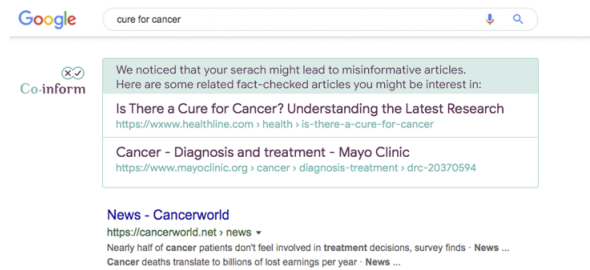


Figure 2: Google Inject presents relevant fact-checked articles before the results, if the query is likely to lead to misinformation

5 FUTURE WORK

Our ongoing analysis of the concepts that emerged from the study, aims at disentangling the design space of nudges for misinformation - the design dimensions against which nudges can be differentiated, as well as the factors that influence their effectiveness and their acceptance. For instance, one can differentiate between *active* nudges (*e.g.*, employing explicit warnings, alerts that prompt reflective choices) and *passive* nudges (*e.g.*, subtle interventions

Table 3: Nudges combating the consumption of misinformation

Description	Bias (Mechanism)
<i>Skim</i> infers users' reading behavior from the mouse (position and speed) and highlights parts that were most likely ignored, when users are exiting a site.	Confirmation bias (Provide multiple viewpoints)
<i>Mitigate</i> identifies queries that most likely lead up to misinformation and automatically suggests alternative search keywords.	Confirmation bias (Provide multiple viewpoints)
<i>Precious</i> infuses fact-checked articles in users' feed, available for reading, only for a limited time period.	Scarcity bias (Make resources scarce)
<i>Google Inject</i> provides recommended articles before the Google results, prompting users to visit fact-checked content (see fig. 2).	Confirmation bias (Provide multiple viewpoints)
<i>Inject</i> identifies topics that users might be susceptible to misinformation and injects related fact-checked articles.	Confirmation bias (Provide multiple viewpoints)
<i>Sorted</i> reorders the results from a search engine query, in a descending order, based on misinformation likelihood.	Status-quo bias (Positioning)
<i>Timing</i> slows down the loading of websites that have been flagged for misinformation.	Peak-end rule (Biasing the memory of past experiences)
<i>Inoculate</i> adds non-credible posts in users' feed and prompts users to visit verified articles they might have not considered before.	Inoculation Theory
<i>Ribbon</i> introduces fact-checked posts on controversial (polarizing) topics on a ribbon, at the top of a social media page.	Continued Influence Effect
<i>Social Translucence</i> notifies of accounts that spread misinformation, while commending users who generate truthful content.	Social Translucence Theory

such as color feedback, as in the case of *Aurole*, or textual feedback, that alerts but doesn't interrupt users) [7]. Passive notifications are more likely to be ignored, disregarded or mistrusted by users [34]. Alternatively, one can differentiate nudges in terms of their time of placement: before, during or after users have viewed misinformation. Empirical research has demonstrated that each strategy - prebunking, labelling, debunking - provides different strengths for prompting behavior change [3]. Also, we cannot overlook ethical concerns and their relation to our concepts. Oftentimes, the debate surrounding nudge ethicality fixates on transparency (*i.e.*, if users can perceive the nudge intention and means of behavior change) and autonomy (*i.e.*, preserving freedom of choice while influencing decision-making) [35]. To address ethical implications, we classify our concepts using a two-dimensional framework proposed by Hansen and Jespersen [16], differentiating nudges in terms of *transparency* and *mode of thinking* engaged, allowing us to distinguish concepts that respect or violate users' autonomy and examine ways of enhancing their morality.

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A APPENDICES

A.1 Design brief

At the beginning of the study, participants were handed printed instructions that included the **"nudge"** definition, along with two examples of such technologies and were also provided with the following design brief:

"Distinguishing between fake news and true information is, at times, challenging. If we do not know what is real and what is fake, we may end up contributing to the dissemination of misinformative content. Misinformation is incorrect and misleading information that is aimed at influencing the public opinion or obscuring the truth. For instance, a news article that had a huge social media engagement, even though it was completely inaccurate, stated that the American president Donald Trump would end school shootings by banning schools. Moreover, a fake news article, which had enormous Facebook engagement, declared that North Korea would open its doors to Christianity, despite the fact that all religions are de facto prohibited. Given the figure below as an example (see fig. 3), imagine that there are two articles for a topic, a fake and a fact-checked article. Twitter users will most likely come across these two pieces of information.

In the first scenario, assume that users encounter a "fake news" article. Your goal is to design a browser plugin for the Twitter platform, which nudges users away from undesired behaviors such as:

- Posting a tweet referring to the fake news article
- Retweeting the fake news article
- Engaging with (i.e., reading) the fake news article

In the second scenario, assume that Twitter users come across the fake news article and that we have a link to a fact-checked article that debunks the ideas proposed in the fake news article. In this case, design a Twitter browser plugin that will nudge users to:

- Engage with the fact-checked article when they encounter the fake news article
- Retweet the fact-checked article or comment on the fake news article (i.e., questioning)

You may develop ideas for any of the two scenarios and for any of the A, B, C directions. Please note that these scenarios and directions are only suggestive; you may develop solutions that go beyond the scenarios and the directions we outlined above."

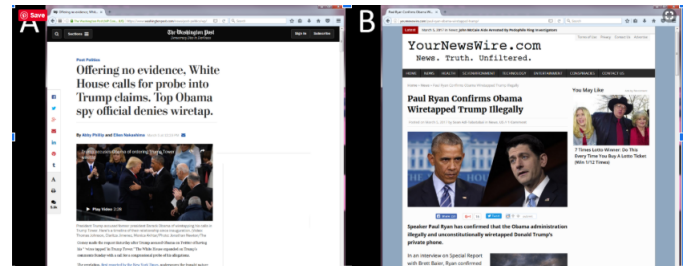


Figure 3: Truthful vs Misinformative news

A.2 Template for designing nudging ideas

this part is removed
after the experiment

Team:

Condition:

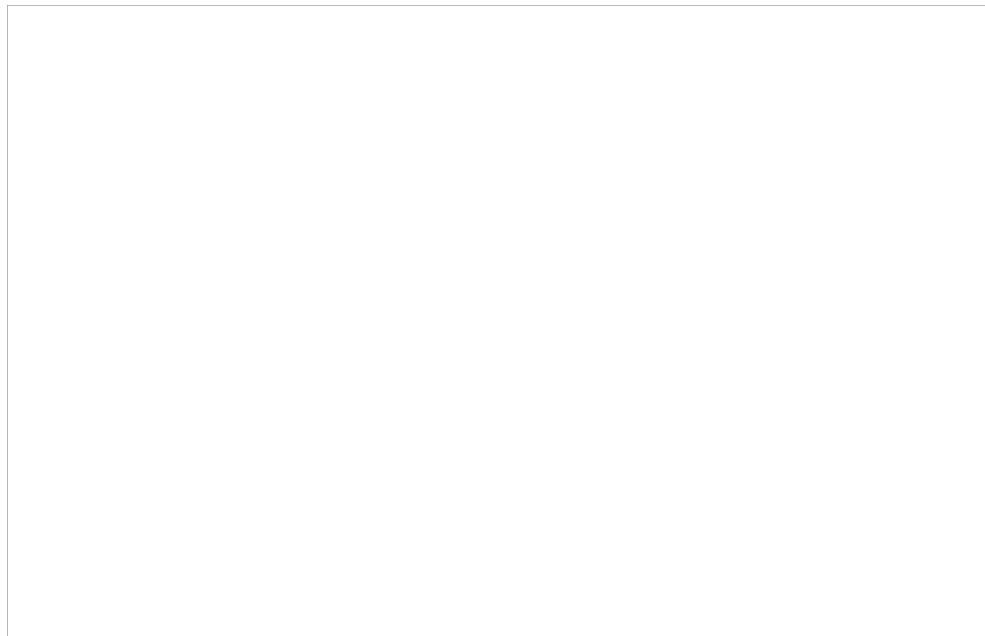


Design Exercise

this part is given to the experts that will evaluate it

Sketch your solutions:

Multiple templates per team





Briefly explain the rationale behind your design(s) solutions:
(e.g., the mechanism(s) used, the technology, the strengths and weak points)

A large, empty rectangular box with a thin black border, intended for the user to write the rationale behind their design solutions.

Further comments:
(e.g., ideas/suggestions or struggles your team faced)

A large, empty rectangular box with a thin black border, intended for the user to provide further comments, ideas, or suggestions.